

COS40007 Artificial Intelligence Engineering

Applied Project Task 1: System Blueprint & Project Planning

Date: 15 April 2026

- Studio Session No. 2
- Group No. 03
- Name and Students' Numbers of Group Members

<i>Name</i>	<i>Student ID</i>
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Chanuli Wijayathilaka	104174006
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1. Group Agreement & Project Plan

1.1 Group style and roles

Our group will work collaboratively by assigning specific roles to each member, while supporting each other throughout the project. Each member is responsible for. A key area, but all members will contribute to discussions reviewing work and ensuring overall quality of the project.

Student name	Roles	Responsibilities
Abbey Stizza	ML engineer	Responsible for model development, including feature engineering, training, testing, and evaluation of the machine learning model.
Amavi uththara	Data engineer	Handles data collection, data understanding, preprocessing, and dataset preparation Also communicates with the tutor on behalf of the group.
Chanuli wijayathilaka	Scrum master	oversees the team during every sprint, plans meetings, makes ensuring work is finished on schedule, and is in charge of gathering, examining, and delivering final deliverables.
Maryam Zaman	Documentation and project coordinator	Manages documentation, GitHub updates, and overall project organisation, ensuring all files are structured and up to date.

1.2 Timeline

Sprint 1: Data preparation (Week 5)

In this sprint, will focus on data understanding and preparing it for the model development.

Tasks:

- Define the problem and the project goal
- Perform Data understanding and data cleaning
- Exploratory data analysis
- data labelling

Sprint 2: Model development (Week 6 – 9)

In this part of the sprint, the focus is on developing the machine learning model

Tasks:

- Feature Engineering
- Split data set in to train/test and validation
- Model technique selection
- Training and hyperparameter setting

- Model Development and testing
- Model optimization

Sprint 3: Deployment and monitoring (Week 10&11)

In this part, the focus is to prepare the model for the practical use.

1.3 Communication Plan

- The main communication platform method for the group will be Discord, for all discussions, files and group calls to occur on the same platform. This will help reduce the chance of information being missed within the group as it's all in one easily accessible location.
- Meetings will be held online once a week. These meetings aim to allow the group to discuss progress and resolve any issues that may arise.
- Shared documents will be managed using OneDrive and Discord.

1.4 On-track behaviours

- Completing tasks on time
- Active participation in meetings
- Clear communication
- Supporting team members

1.5 Off-track behaviours

- Missing deadlines
- Lack of communication
- Not contributing to assigned tasks
- Leaving assigned sections until the last minute

1.6 Resolving tensions

Conflicts will first be discussed within the group. If unresolved, the tutor will be consulted for guidance.

2. System Blueprint Diagram

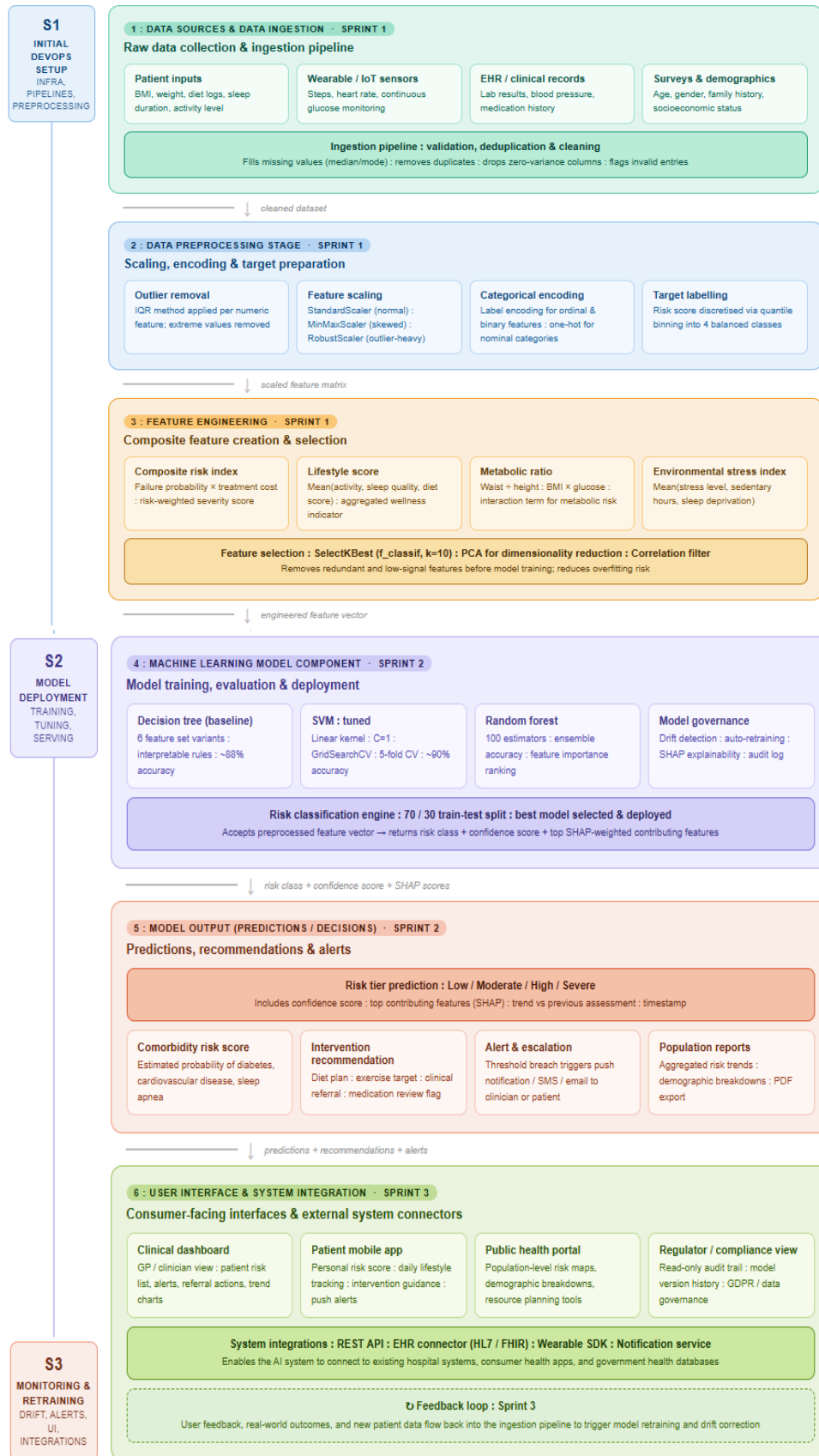


Figure 1: Architecture design for health risk monitoring AI system

The architecture diagram shows how the Health Risk Monitoring AI System is structured across six stages and three development sprints. Data is collected from four sources; patient inputs, wearables, clinical records, and surveys. It is then passed through an ingestion pipeline that cleans and validates it. The cleaned data is pre-processed through scaling and encoding, before engineered composite features are created to improve model signal. The prepared data feeds into the machine learning component, where three models are trained and compared, with the best deployed as the risk classification engine. The model outputs a risk tier (Low / Moderate / High / Severe), comorbidity scores, and intervention recommendations, which are delivered to end users through a clinical dashboard, patient mobile app, public health portal, and regulator view. A feedback loop in Sprint 3 returns real-world outcomes back into the pipeline to support ongoing model monitoring and retraining.

3. MLOps & DevOps Integration

3.1 Data & Training Strategy

The dataset consists of 2111 instances and 16 features, which will be divided into three parts: training, validation, and testing, using a 70:15:15 split. The training set (70%) will be used to train the machine learning model. The validation set (15%) will be used to tune hyperparameters, compare different models, and detect overfitting. The remaining testing set (15%) will be used to evaluate the final model performance on unseen data and estimate how the model will perform in real-world situations. This data and training strategy ensures that the model is reliable and accurate when applied to real-world data.

New data will be used for retraining. As patients use the system, their updated readings, outcomes and feedback are collected and then fed back into the ingestion pipeline. This gives the model fresh examples and data to learn from as individual's lifestyle patterns change over time.

3.2 Retraining strategy

The model retraining strategy is designed to ensure that the Health Risk Monitoring AI System remains accurate and relevant over time and requires minimal manual training [1]. Retraining will be triggered using both performance-based and time-based approaches. Performance-based retraining will occur when there is a significant decrease in model performance such as a decrease in accuracy, e.g. by 10%, that indicates the model is not producing reliable data. Time-based retraining will occur on a set schedule e.g. monthly, to include updated data and ensure the system is up to date [2]. Retraining may also be triggered when a data drift is detected. When retraining is triggered, the input data collected would go through the preprocessing and feature engineering stages. This would occur before the model is retrained and updated. Combining these approaches allows the model to develop with time while maintaining its reliability and performance accuracy.

3.3 Monitoring & Drift Detection

There are two main types of drift that need to be considered, data drift and content drift. Data drift occurs when the distribution of incoming patient features changes. For example, if the system begins receiving a high volume of data from an older demographic that it is not trained on. This is detected by comparing the statistical data (e.g. mean, standard deviation, distribution) of new inputs against the training data baseline.

Content drift occurs when the relationship between the features and its correlated result changes. For example, if new guidelines are released which redefine a 'high risk' BMI, the results from the model would be inaccurate. Concept drift is harder to detect than data drift and would usually be picked up through a drop in model accuracy or if it was flagged by clinical reviewers.

When the system encounters either type of drift, the pipeline pulls in the newly accumulated data and reprocesses it through the preprocessing and feature engineering stages. It then retrains the model, evaluates it against a holdout set (a portion of data purposefully withheld during the model training), and if the performance improves, it promotes the new model to production. The previous iteration of the model is archived for ease of traceability. This workflow ensures the system is relevant and adapted to real-world data.

3.4 DevOps Practices

GitHub will be primary control platform used by the group. As a team, we will be able to store codes, monitor changes and collaborate more successfully as a result. It is helpful because team members may work on different parts of the project, and GitHub facilitates the structured integration of such modifications.

Continuous Integration and Continuous Deployment (CI/CD) is a collection of techniques and resources intended to enhance the software development process by automating builds, testing, and deployment, allowing to deliver code modifications more quickly and consistently. These techniques can increase productivity by automatically testing data before merging, reducing errors, and ensuring verified changes are implemented, thus minimising manual labour [3].

In machine learning, model versioning is crucial as models are built using both data and code. Depending on the settings and data used, each trained model could result in various outcomes. As a result, several iterations of the trained models will be stored and monitored throughout time in the project.

4. Project Management Setup

4.1 Agile project management tool

Our group is using Trello as the main Agile project management tool to organise tasks, assign responsibilities, monitor progress, and manage deadlines throughout this project. The board is structured into five workflow stages: Backlog, To Do, In Progress, Review, and Done. This

setup helps the team track work clearly and maintain accountability during each stage of development.

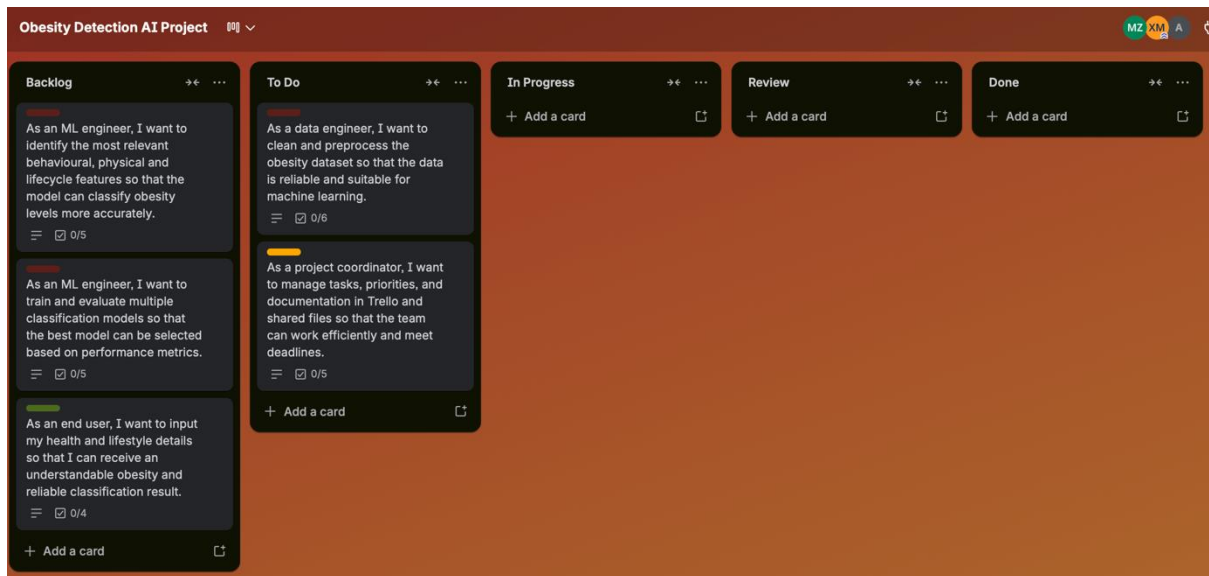


Figure 2: Workspace created by Trello

4.2 Internal user stories

Figure 2 shows the below user stories in the groups Trello workspace.

1. Data Preprocessing: As a data engineer, I want to clean and preprocess the obesity dataset so that the data is reliable and suitable for machine learning.
2. As an ML engineer, I want to identify the most relevant behavioural, physical and lifestyle features so that the model can classify obesity levels more accurately.
3. As an ML engineer, I want to train and evaluate multiple classification models so that the best model can be selected based on performance metrics.
4. As a project coordinator, I want to manage tasks, priorities, and documentation in Trello and shared files so that the team can work efficiently and meet deadlines.
5. As an end user, I want to input my health and lifestyle details so that I can receive an understandable and reliable obesity classification result.

4.3 Break user stories into tasks, assign responsibilities and create checklists.

User Story 1: Data preprocessing

Assigned to: Amavi uththara

Priority: High

Tasks: inspect dataset structure, check missing values, check duplicates, encode categorical variables, scale numerical variables, split dataset into training, validation, and testing sets.

User Story 2: Feature selection

Assigned to: Abbey Stizza

Priority: High

Tasks: review dataset variables, identify important obesity-related features, check feature correlations, remove redundant features if required, prepare the final feature set.

User Story 3: Model development

Assigned to: Abbey Stizza

Priority: High

Tasks: select baseline classification models, train initial models, tune hyperparameters, evaluate model performance, compare accuracy, precision, recall, and F1-score, and select the final model.

User Story 4: Documentation and coordination

Assigned to: Maryam Zaman

Priority: Medium

Tasks: write the project description, write the system overview, record key design decisions, update meeting notes, and prepare the contribution statement.

User Story 5: Prediction workflow

Assigned to: Chanuli Wijayathilaka

Priority: Medium

Tasks: define input variables, define output obesity classes, describe the prediction workflow, and align the output with the system blueprint.

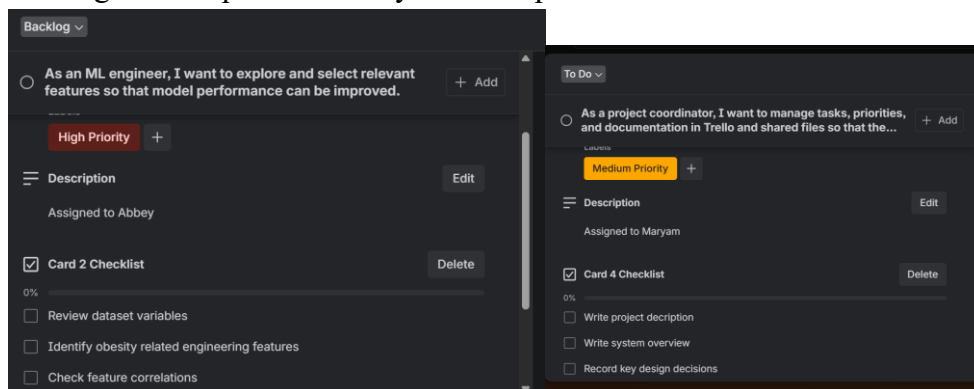


Figure 3,4 – priority listing and checklists for each user story.

4.4 Documentation

Project Description

This project applies machine learning techniques to classify and predict obesity levels using behavioural, physical, and lifestyle-related attributes. The obesity dataset from Studio 1 estimates obesity levels based on eating habits and physical condition. The aim is to develop a clear and structured AI system that supports data preparation, feature engineering, model

development, and prediction results in a way that can be managed across multiple development sprints.

System Overview

The proposed system uses the obesity dataset as its primary data source which provides lifestyle health inputs. The dataset is first ingested and prepared through preprocessing steps such as cleaning, encoding categorical variables, and scaling numerical features. Relevant behavioural and physical features are then selected and used to train and evaluate multiple machine learning classification models. The final system is designed to output a predicted obesity category (Low / Moderate / High / Severe) based on the user's input variables, working as a health monitoring system. The workflow is aligned to the project's sprint structure and Trello task management setup.

Key Design Decisions

The obesity dataset from Studio 1 was selected as the foundation of the system as it is directly relevant to the chosen problem domain and suitable for classification modelling. The dataset will be split into training, validation, and testing sets using a 70:15:15 ratio to support model development, tuning, and fair evaluation. Multiple classification models will be trained and compared before selecting the final model based on performance metrics. Trello was chosen as the main Agile project management tool for tracking user stories, task assignments, and priorities, while GitHub will be used for version control and later DevOps integration.

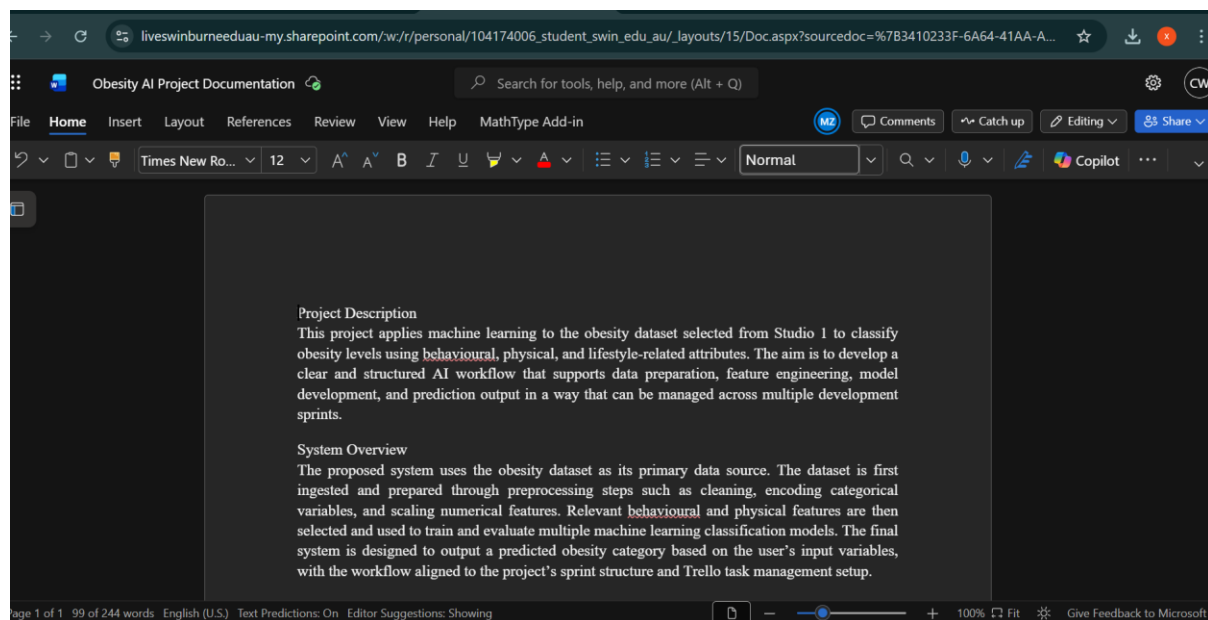


Figure 5 – Shared word documentation

- Trello Link
<https://trello.com/invite/b/69df1cc19301edfcbdc1f9a/ATTI1b260ff31a599e7b17955689b500176eC8AEFC75/obesity-detection-ai-project>

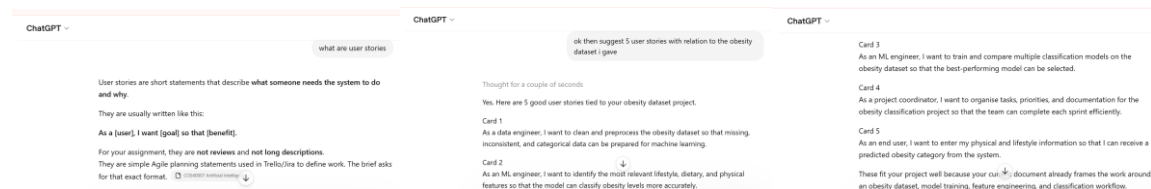
- Shared Word File Link
[Obesity AI Project Documentation.docx](#)
- GitHub Link
https://github.com/COS40007-2026-Classrooms/CL02_G03

5. Contribution Statements

- Abbey Stizza – completed section 2 “System Blueprint Diagram” and completed section 3 , “3.3 Monitoring & Drift Detection” subsection.
- Amavi Uththara – completed section 1 “Group Agreement & Project Plan” and section 3, “3.4 DevOps Practices” & “3.1Data & Training Strategy-How will data be split” subsections.
- Chanuli Wijayathilaka – Completed section 4 ‘Project Management Setup’ including Trello setup and the shared document.
- Maryam Zaman – completed section 3.2 “Retraining Strategy”, assisted with the completion of other section 3 tasks, formatted and submitted document.

6. GenAI Declaration

Chanuli: ChatGPT was used to write the user stories as mentioned in the studio 5 document.



6. References

- [1] “What is Machine Learning Model Retraining?,” Iguazio, <https://www.iguazio.com/glossary/model-retraining/>
- [2] S. J. Basa, “Mastering model retraining in Mlops | RandomTrees, Data & AI Innovations Company |,” Medium, <https://randomtrees.medium.com/mastering-model-retraining-in-mlops-4bb961ec7070>
- [3] S. Pittet, “Continuous integration vs. continuous delivery vs. continuous deployment,” Atlassian, 2019. <https://www.atlassian.com/continuous-delivery/principles/continuous-integration-vs-delivery-vs-deployment>
- [4] Harry Connor AI, “Train, Validation & Test Sets in Machine Learning,” YouTube, Feb. 06, 2026. <https://www.youtube.com/watch?v=ZQ4IwBg6U5c>
- [5] GitHub, “What is CI/CD?,” *GitHub*, Nov. 07, 2024. <https://github.com/resources/articles/ci-cd>
- [6] “Versioning Code and Models | Outerbounds,” *Outerbounds*, 2026. <https://docs.outerbounds.com/versioning-code-and-models/> (accessed Apr. 15, 2026).